

CELLULAR INSERTION OF MEMBRANE PROTEINS FOR THEIR STRUCTURAL ANALYSIS USING NMR SPECTROSCOPY

**Noé IMLER¹, Cléo BONILLO¹, Morgane CALLON¹, Marie-Laure FOGERON¹, Anja
BÖCKMANN¹**

*¹Molecular Microbiology and Structural Biochemistry (MMSB), UMR 5086 CNRS-Université de Lyon, 69367
Lyon, France
noe.imler@ibcp.fr*

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ABSTRACT

Solid-state NMR allows the study of insoluble proteins as membrane proteins. These membrane proteins are generally studied in a lipid native-like environment. One problem that remains today is how to study them directly in cells, to identify the impact of partner proteins and structural modifications induced by the cellular environment. In this context, nanodisc-based approaches allow the insertion of membrane proteins labeled for NMR, and thereby provide a way for the structural study of membrane proteins inside cells.

The aim of this work is to establish the insertion of membrane proteins into cells, enabling NMR measurements in a more native-like environment.

Our work initially focuses on the cell insertion of the model protein OmpX, produced in bacteria or with a wheat germ cell-free protein synthesis, using a nanodisc system. Best settings for optimising insertion are being evaluated. The results are analyzed by fluorescence microscopy, FRET and spectral flow cytometry.

Two different nanodiscs compositions were evaluated yet for insertion into a membrane bilayer model and eukaryotic cells using different incubation times and nanodiscs concentrations.

This nanodisc-mediated cell insertion of membrane proteins shall pave the way for the structural characterization of viral membrane proteins directly in cells.

REFERENCES

1. Umbach S. et al. *Front. Bioeng. Biotechnol.* 10, 906295 (2022). DOI: 10.3389/fbioe.2022.906295
2. Patriarchi T. et al. *Sci Rep* 8, 3556 (2018). DOI: <https://doi.org/10.1038/s41598-018-21863-3>